

		amount of substance in the system. $U = \frac{3}{2}NkT$
13	C	Using $g = GM/r^2$ and $\rho = \text{mass} / \text{volume} = M / [(4/3)\pi r^3]$, $g = G[\rho(4/3)\pi r^3] / r^2$ $= (4/3)G\rho\pi r$ $g_E/g_M = \rho_E r_E / \rho_M r_M = \rho_E r_E / \rho_M r_M$ $6 = (5/3) / (r_E / r_M)$ $r_E / r_M = 3.6$
14	C	Change in GPE